TEER 文獻雙週回顧

2026-03-11 至 2026-03-25

謝慕揚 MD, PhD, FESC | 2026-03-25

29 篇文獻 | PubMed & Semantic Scholar

Key Pearls (1/2)

Pearl 1: Residual **LA v wave ≥25 mmHg** after M-TEER 獨立預測 death/HF hospitalization，提供超越 echo MR grading 的增量預後資訊

Pearl 2: M-TEER 長期死亡主因為 HF (34.6%)，**TAPSE/sPAP ≤0.36** 和 **≥moderate TR** 是獨立預測因子；**MitraScore** 在 8+ 年追蹤仍有效

Pearl 3: T-TEER vs surgical TVr (Medicare, n=2,761)：手術住院 mortality 較高 (OR 2.83) 但 **3 年 survival 較佳** (HR 0.76)

Key Pearls (2/2)

Pearl 4: APSC 亞太共識 — 16 項 TEER 聲明 (TR + MR)，台灣宋思賢醫師為共同作者

Pearl 5: AI fusion imaging (DeviceGuide) 自動追蹤 PASCAL Ace 並對齊 3D TEE/fluoroscopy，51 例多中心經驗改善團隊溝通

Pearl 6: Baseline **VAC (TAPSE/PASP) ≤0.39** 在 T-TEER 術前可預測不良預後

Mitral TEER

Long-Term Mortality After M-TEER

Popolo Rubbio et al. *Cardiovasc Revasc Med.* 2026.

- 218 例 MitraClip (2016-2020)，中位追蹤 3.3 年 (最長 8.5 年)
- 130 例 (59.6%) 死亡：**CV 55.4%** (HF 34.6%)、non-CV 44.6%
- 存活者 (n=88)：僅 12.5% HF 再住院，2.3% repeat M-TEER

獨立預測因子	
Ischemic secondary MR etiology	TAPSE/sPAP ≤0.36
Prior heart failure	≥Moderate TR
Low MitraScore → better survival	High: 83.9% vs Low: 48.6% mortality

Residual LA v Wave Predicts Outcome

Paulus et al. *ESC Heart Fail.* 2026.

- 前瞻性 299 例 M-TEER，2 年追蹤
- 術後 LAvP: 30.5 → **23.2 mmHg** ($p<0.001$)
- **LAvP per 10 mmHg ↑: HR 1.37** (95% CI 1.15-1.63, $p<0.001$)
- LAvP <25 mmHg → favorable outcome，即使 residual MR grade I-II

> **Take-home**: 術中 LA pressure monitoring 應輔助 echo assessment，LAvP <25 mmHg 為理想目標

M-TEER in Advanced Heart Failure

Crivelaro et al. *Catheter Cardiovasc Interv*. 2026.

- 13 例 advanced HF (median LVEF **22%**, EuroSCORE II **10.4%**)
- 38% 需 continuous inotropic support
- **100% procedural success**，無 30-day mortality
- 80% inotrope patients 成功 weaning
- 1 年 61.5% 無 composite endpoint

> M-TEER 可作為 advanced HF 的 bridging strategy

M-TEER in Very Elderly (EXPAND Studies)

Price et al. *J Am Heart Assoc.* 2026.

- **854 例 primary MR**，最年長 Q4: 86-99 歲
- 30-day MAE: 5.7% vs 4.3% (NS)
- 所有年齡組均有顯著改善：
  - LV reverse remodeling
  - NYHA functional class
  - KCCQ-OS quality of life

> **Age alone should NOT be a barrier** to M-TEER with next-generation devices

Predictors of Non-Response in AFMR

Kandiah et al. *Can J Cardiol*. 2026. (Review)

- AFMR 為獨特的 MR 亞型：atrial remodeling + annular dilatation
- Non-response 預測因子：
  - Atriogenic leaflet tethering
  - RV dysfunction
  - Residual post-procedural MR
  - Elevated mean transmitral gradient
- 強調 AFMR 患者選擇和 procedural planning 的最佳化

FED vs Barlow Disease in DMR TEER

Barki et al. *EHJ Cardiovasc Imaging*. 2026.

	FED (n=91)	Barlow (n=34)	p
Mean clips	1.53±0.6	**1.82±0.6**	0.01
≥2 clips	46%	**76%**	0.003
30-day success	88%	85%	0.76
12-mo event-free	**Higher**	Lower	**0.03**

- Barlow disease: 可行且安全但需更多 clips
- **FED 有更好的 12 個月 event-free survival** (HR 3.13)

Tricuspid TEER

T-TEER vs Surgical TVr (Medicare)

Mehaffey et al. *Ann Thorac Surg*. 2026.

	T-TEER (n=1,540)	Surgery (n=1,221)
Hospital mortality	2.1%	8.8%
Adjusted hospital mortality	Reference	OR 2.83
3-year survival	Reference	HR 0.76 (p=0.012)
3-yr HF readmission	Reference	HR 0.79 (p=0.001)
3-yr reintervention	Reference	HR 0.83 (p=0.003)

> Surgery: higher perioperative risk but **better long-term outcomes**

T-TEER vs TTVR (Cardiovalve System)

Marquetand et al. *Clin Res Cardiol*. 2026.

- T-TEER 104 例 vs TTVR 10 例 (anatomical suitability)
- 30 天 TR ≤ grade I：T-TEER **44.9%** vs TTVR **80%** ($p<0.001$)
- NYHA I/II 改善相似 (63.2% vs 70%)
- TTVR major bleeding 較高 (20% vs 1.96%, $p=0.041$)
- 無死亡、stroke 或 surgical conversion

> TTVR 達到更好的 TR reduction，但 T-TEER 安全性更佳

Survival Benefit with High TRI-SCORE

Gröger et al. *Eur J Heart Fail*. 2026. (EuroTR Registry)

- 多中心 EuroTR registry 資料
- 高 TRI-SCORE 患者在 successful T-TEER 後仍有 survival benefit
- 支持 T-TEER 在高風險患者中的效益

RV-Arterial Coupling After T-TEER

Angulo Lara et al. *EHJ Cardiovasc Imaging*. 2026.

- 62 例 T-TEER (TriClip 60%, Pascal 30%)
- Baseline **TAPSE/PASP ≤0.39** → higher death/HF admission (p=0.038)
- T-TEER 後 sPAP 顯著降低，RV function 穩定
- VAC 改善在無 events 患者更明顯

> VAC 提供比傳統 RV function 參數更精確的預後資訊

Liver Stiffness Reversibility After T-TEER

Angulo Lara et al. *EHJ Cardiovasc Imaging*. 2026.

- 34 例 T-TEER，liver elastography 追蹤
- Liver stiffness: **8.9 → 6.7 kPa** at 1 year (p=0.013)
- Fibrosis stage: **F2 → F1** (p=0.003)
- F0/F1 比例: 53% → **70%** at 1 year

> 較高 liver stiffness **不應視為 T-TEER 禁忌症**，成功 TR correction 可逆轉 liver fibrosis

Disease Stage Impact on T-TEER

Velarde Morales et al. *EHJ Cardiovasc Imaging*. 2026.

- 44 例 Pascal device: early-stage (27%) vs intermediate-advanced (73%)
- Procedural success 相似 (~95%)
- **Early-stage event-free survival 顯著優於 intermediate-advanced** (p=0.04)

> 支持 **earlier intervention** 概念 — 等太久可能失去最佳治療窗口

T-TEER in Asian Patients

Chan et al. *JACC Asia*. 2026.

- 香港單中心回顧性研究
- TriClip 或 MitraClip "miskey" technique
- 亞洲人群 T-TEER 早期臨床及超音波結果
- 為亞太地區 T-TEER 實踐提供初步數據

Imaging / Technique / Guidelines

AI Fusion Imaging for M-TEER

Biaggi et al. *Eur Heart J.* 2026.

- **DeviceGuide EchoNavigator SmartVue** (Philips)
- AI 自動追蹤 PASCAL Ace + 對齊 3D TEE/fluoroscopy
- 多中心 51 例，4 個 structural heart centers (US/NL/CH)
- **Target Mode**: trajectory overlays for navigation
- **Device Mode**: auto-centred device visualization
- 團隊評價：helpful/very helpful，改善溝通與策略討論

SORI Phenomenon During TEER

Lin et al. *Catheter Cardiovasc Interv*. 2026.

- 首次命名 **"Small Orifice Regurgitation Increasing" (SORI)** 現象
- Commissural MR 在 TEER 術後可偵測到比術中更高等級的 MR
- 2 例 commissural MR case series
- 對理解和預防 residual commissural MR 有重要參考價值

> Commissural MR 行 TEER 時須警覺 SORI 現象

APSC TEER Consensus Statements

Yeo et al. *J Asian Pac Soc Cardiol*. 2026.

- 更新 2021 年 APSC 共識 (原僅限 MitraClip for MR)
- **16 項共識聲明**：涵蓋 TR 和 MR 的 TEER
- GRADE 系統評估證據，≥80% 投票同意
- 新增 PASCAL、TriClip 裝置的建議
- **台灣共同作者：宋思賢 (Shih-Hsien Sung)**

> 亞太地區 TEER 實務的重要參考文件

2025 ESC/EACTS VHD Guidelines Update

Patel et al. *Curr Opin Cardiol*. 2026. (Review)

- Imaging-based staging of cardiac damage 正式化
- 擴大 mitral & tricuspid TEER 適應症
- 降低 AS 和 primary MR 的 intervention thresholds
- 強調 Heart Valve Centres 和 volume-outcome relationships
- 心臟外科 + 心臟內科 partnership 至關重要

Case Reports of Interest

MitraClip Embolization → Pulmonary Vein

Kumi et al. *JACC Case Rep*. 2026.

- 77 歲男性 cardiogenic shock with severe MR
- 第二顆 MitraClip 植入時 **locking threads knotting**
- Clip embolization 至 **right lower pulmonary vein** (首例報告此位置)
- 使用 gooseneck snare through multipurpose catheter 成功取出

> **Learning**: PV 是罕見但可能的 embolization 位置；適當的 catheter 選擇和 grasp orientation 是安全取出的關鍵

Leaflet Fragment → Coronary Occlusion (Fatal)

Kosaki et al. *Can J Cardiol*. 2026.

- Rescue TEER for acute severe MR，**repeated leaflet grasping**
- Fragile leaflet perforation → 碎裂的 native mitral leaflet tissue embolization
- 導致 **acute coronary artery occlusion**
- Autopsy 確認 embolic material 為 mitral leaflet tissue
- 患者死亡

> **Warning**: 此為 TEER 中未被認識的 coronary obstruction 機制；fragile leaflet 須格外小心

TEER in Massive Left Atrium

Ni et al. *EHJ Case Rep*. 2026.

- 77 歲男性，**giant LA (113×129×133 mm)**，severe MR
- Tailored **posteroinferior transseptal puncture** (4.67 cm above annular plane)
- 成功植入 **3 顆 MitraClip**，mean gradient 5 mmHg，residual mild MR
- 1 個月 LA reverse remodeling 及症狀改善

> **Learning**: Optimized TSP 是成功關鍵；不應受限於 rigid anatomic thresholds

Mavacamten + M-TEER for oHCM

Wang et al. *JACC Case Rep*. 2026.

- 79 歲女性，severe oHCM + MVP + AS + secondary TR
- **分步策略**：先 mavacamten → 再 M-TEER
- 顯著改善症狀和血流動力學
- 直接 M-TEER 在 severe LVOT obstruction 下風險較高

> **Learning**: oHCM 中辨別 MR 機轉至關重要；mavacamten 可先降低 LVOT gradient 再行 M-TEER

Bail-Out M-TEER After Failed NeoChord

Putti et al. *Am J Cardiol*. 2026.

- 79 歲男性 severe DMR (P2 flail) 行 NeoChord 修補
- 術後 **NeoChord apical detachment** → recurrent severe MR + new LV dysfunction
- 需 **VA-ECMO + CRRT**，禁忌 redo surgery
- Urgent M-TEER with VA-ECMO support：**2 顆 MitraClip XTW G4** 成功

> **Learning**: M-TEER 可作為 surgical repair 失敗後的 bail-out strategy

TEER for Papillary Muscle Rupture / Dextrocardia

其他值得關注的 Case Reports

TEER for papillary muscle rupture after AMI

Chen et al. *中華心血管病雜誌*. 2026.

- 回顧 TEER 治療 AMI 後 papillary muscle rupture 的中國經驗

MitraClip in mirror-image dextrocardia

Li et al. *Catheter Cardiovasc Interv*. 2026.

- 中國首例 dextrocardia + MR 行 MitraClip；解剖完全反轉的技術挑戰

本期文獻摘要

類別	篇數	重要主題
Mitral TEER	8	LA v wave、advanced HF、elderly、AFMR、FED vs Barlow
Tricuspid TEER	8	Surgery comparison、VAC、liver stiffness、disease staging、Asian data
Imaging/Technique	4	AI fusion、SORI phenomenon、CT annular dimensions
Guidelines	2	APSC consensus、ESC/EACTS VHD update
Case Reports	7	Embolization、massive LA、oHCM、NeoChord failure
Total	29	

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謝謝聆聽

Q & A

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TEER Biweekly Review | 2026-03-25